Yilin Wang

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Yilin Wang | Dylan-Wyl10 | 0000-0002-6002-0535

West Lafayette, IN - 47906, USA

Research Interests

Seeking **Research Scientist / Applied Scientist / ML Engineer** roles in intelligent transportation systems and autonomous driving:

- **Robotaxi & AV fleet operations** – routing, dispatching, and repositioning at network scale, with fleets doubling as mobile sensing platforms; via MILP and learning-based combinatorial optimization.
- **LLM/VLM agents & world models for AV & physical AI** – mobility planning foundation models for sim agents, world-model construction, and safety-critical scenario generation & closed-loop validation.
- **Cooperative perception & V2X systems** – from algorithms to real-world field deployment.
- **Agentic system implementation & deployment** – multi-agent architectures, tool orchestration, and agent evaluation for production workflows, from smart-city & robotaxi operations to automated research pipelines.

Education

- **Purdue University** *Aug.2021 - May 2027 (Expected)*
  Doctor of Philosophy in Civil Engineering (Transportation and Infrastructure Systems Engineering) West Lafayette, US
  - Research Area: Connected and Automated Vehicle, Cooperative Perception, Physics-informed Machine Learning, Robotaxi Operation and Control
  - GPA: 3.8/4.00
- **Carnegie Mellon University** *Aug.2019 - Dec.2020*
  Master of Science in Advanced Infrastructure Systems Pittsburgh, US
  - Area: Smart City, Deep Learning, Natural Language Processing
  - Course: Deep Learning, Reinforcement Learning, Geo-Informatics System
  - GPA: 3.87/4.00
- **Tongji University** *Aug.2015 - Jun.2019*
  Bachelor of Science Shanghai, China
  - Major: Civil Engineering (Area of Geo-technical and Underground Engineering)

Working Experience

- **Inchor(SaferDrive AI)** *Dec.2025 - Jun.2026*
  Simulation Development Engineer Intern Ann Arbor, MI
  - **TeraSim – Naturalistic Driving Behavior**
    - Developed and maintained TeraSim, an open-source city-scale traffic simulation platform for AV safety testing.
    - Led integration of MotionTransformer as the foundation model of NNDE, generating naturalistic driving behavior at city scale alongside the NADE adversarial-scenario module; refactored the behavior-model pipeline from a dual-state design into a stateless prediction pipeline for large-scale inference.
    - Implemented a full-stack Driving Foundation Model into product; led the joint project with NVIDIA and the City of San Jose on agent-driven Smart City operation.
    - For that project, led a city-scale urban simulation demo under the **NVIDIA Metropolis Blueprint**: an integrated "City Operations Agent" and digital twin that perceive, simulate, and optimize traffic operations; designed the full simulation pipeline, including city-scale vehicle injection, dynamic routing, and multi-phase demo orchestration.
    - The demo with NVIDIA and the City of San Jose was **selected for NVIDIA CEO Jensen Huang's GTC 2026 keynote**.
  - **AgentSim – Agentic Synthetic Scenario Generation** [Demo]
    - Provide an end-to-end pipeline: from trajectory/text description to safety-critical scenarios with ego camera-view data.
    - Develop an agentic platform with LLMs/VLM agents to collaborate, reconstruct, mutate, and render driving scenarios.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, S=UNDER REVISION

[S.1] **Yilin Wang**, Yiheng Feng* (2026). **RoboSense: Leveraging Robotaxi Fleets as Drive-by Sensors for Urban Traffic Monitoring**. Under revision at *Transportation Research: Part C*. Preprint: arXiv:2609.06813.

[J.1] Hanlin Chen, Vamsi K Bandaru, **Yilin Wang**, Mario A Romero, Andrew Tarko, Yiheng Feng* (2024). Cooperative Perception System for Aiding Connected and Automated Vehicle Navigation and Improving Safety in *Transportation Research Record*, 2678(12), 1498-1510. https://doi.org/10.1177/03611981241252779

[J.2] **Y. Wang** and Y. Feng* (2024). IDM-Follower: A Model-Informed Deep Learning Method for Car-Following Trajectory Prediction in *IEEE Transactions on Intelligent Vehicles*, vol. 9, no. 6, pp. 5014-5020, June 2024, doi: 10.1109/TIV.2024.3367654

[J.3] Ya-Dong Xue, Wei Zhang, **Yi-Lin Wang***, et al. (2023). Serviceability evaluation of highway tunnels based on data mining and machine learning: A case study of continental United States in *Tunnelling and Underground Space Technology*, Volume 142, 2023, 105418, ISSN 0886-7798, https://doi.org/10.1016/j.tust.2023.105418

[C.1] **Yilin Wang**, Yiheng Feng* (2026). **RoboSense: Leveraging Robotaxi Fleets as Drive-by Sensors for Urban Traffic Monitoring**. In *INFORMS 2026 Annual Meeting*, Accepted Talk.

[C.2] **Yilin Wang**, Yiheng Feng* (2026). A Cooperative Perception Based Dynamic Vehicle Routing Framework for Urban Traffic Monitoring. In *Transportation Research Board Annual Meeting 2026*, Accepted by Poster Presentation.

[C.3] **Yilin Wang**, Yiheng Feng* (2024). Dynamic Routing of Connected and Automated Vehicles for Improving Network Coverage. In *Transportation Research Board Annual Meeting 2024*, Accepted by Poster Presentation.

PROJECTS

- **Cooperative Perception CAV Fleet Operation for Data Collection** *Apr.2023 - Jun.2026*
  Individual Research Project [GitHub]
  - **Project 1**: Dynamic Routing on Connected and Automated Vehicles for Improving Network Coverage
    Apr.2023 - Aug.2024
    - **Research Question**: How to model the CAV network coverage and operate routes to improve data collection.
    - **Contribution**:
      1. We formulate the CAV routing problem considering the network coverage as one objective.
      2. We propose a heuristic algorithms with greedy search to solve the multi-objective optimization.
      3. We conduct a simulation experiment in a grid network to verify the feasibility and relations between travel cost and network coverage.
    - **Methodology**: heuristic optimization, SUMO simulation, link-based travel time estimation.
    - **Outcome**: Accepted by Transportation Research Board (TRBAM) 2024 as poster presentation.
  - **Project 2**: RoboSense: Leveraging Robotaxi Fleets as Drive-by Sensors for Urban Traffic Monitoring
    Sept.2024 - Jun.2026
    - **Research Question**:
      1. A higher granularity of road network representation is needed.
      2. A proper performance metric is needed to evaluate traffic monitoring impact.
      3. There is a dynamic effect between CAV routes decision and partial network observations.
    - **Contribution**:
      1. We apply Cell Transmission Model (CTM) for traffic state prediction and road network representation considering CAV detection maneuver.
      2. We propose a comprehensive MILP formulation considering the performance of traffic monitoring as well as dynamic vehicle routing in one explicit equation.
      3. We conduct a microscopic simulation and evaluations under various scenarios.
    - **Outcome**: Under revision at Transportation Research: Part C; preprint arXiv:2609.06813; poster at TRB 2026; accepted talk at INFORMS 2026.

- **TeraSim: City-Scale World Model for Safety Testing on AV and Physical AI** *Dec.2025 - Present*
  Core Developer | Internship | SaferDrive AI(Inchor)
  - **Overview**: A city-scale simulation platform serving as the foundational infrastructure for agentic world models, enabling naturalistic and adversarial testing of autonomous vehicles at urban scale.
  - **Core Module – NNDE (Neural-network Naturalistic Driving Environment)**: Behavioral-layer foundation module of TeraSim that integrates a transformer-based motion model to drive city-scale, naturalistic AV behavior generation.
  - **Core Module – AgentSim (Agentic Test Engineer)**: An end-to-end, LLM/VLM-driven multi-agent platform that generates safety-critical scenarios with trajectories or text input. [Demo]

- **Milestone**:

  1. 2026-03: Active deployment with **NVIDIA** and **City of San Jose** as an *Agentic Incident-to-Operation* pipeline, bringing TeraSim from simulation into real-world city operations.
  2. 2026-06: First draft release of AgentSim, utilizing LLMs and VLMs for safety-critical scenario generation.
  3. 2026-10: Public AgentSim demo: 3 real drives, each with 5 generated safety-critical variants rendered as synchronized front-camera video.

- **Tech Stack**: Python, SUMO, PyTorch, CUDA, Docker, FastAPI

- **Modeling Car-Following Behavior with Physics-Informed Neural Network** *Jan.2022 - Present*
  Behavior-level reconstruction of human driving: from physics-constrained prediction to cognitively interpretable response [GitHub]
  - **Overview**: Reconstruct human driving behavior by embedding physical and cognitive structure into neural networks, so that learned behavior models stay accurate, robust, and interpretable – the behavior layer behind sim agents and world models.
  - **Project 1**: IDM-Follower: A Model-Informed Deep Learning Method for Car-Following Trajectory Prediction *Jan.2022 - Feb.2024*
    - **Research Question**: How to integrate an explicit, physics-based car-following model into a learning-based predictor to improve training and performance.
    - **Contribution**:
      1. We introduce a physics-informed neural network (PINN) model that integrates the Intelligent Driver Model (IDM) into a neural trajectory predictor.
      2. The proposed model exhibits robustness against real-time GPS noise.
    - **Methodology**: A customized attention-based VAE trained with a loss that combines physical-law residuals and ground-truth error.
    - **Outcome**: Published in **IEEE Transactions on Intelligent Vehicles** (first author); research presentation at IEEE-IV 2024, Jeju Island, KR.
  - **Project 2**: Neural Force: Interpretable Learned Car-Following Response Fields *2024 - Present*
    - **Research Question**: How does a driver combine perceptual cues – spacing, closing speed, and own speed – into one acceleration decision, and can a learned model keep that combination readable instead of hiding it in a black box?
    - **Contribution**:
      1. We propose a cognitively motivated response-field model: acceleration is assembled from signed, weighted gradients of one shared learned scalar, and each term carries a behavioral role (restore spacing, oppose closing, damp speed).
      2. We derive interpretable readouts – operating gap, gap-restoration gain, and closing-opposition gain – that make learned human behavior comparable across datasets.
      3. Across four naturalistic corpora (AD4CHE, highD, LA Highways, I-24 MOTION), we find a shared corrective structure but distinct sensitivities: five of six corpus pairs reverse order between the two gains, ruling out a single "response strength" axis.
    - **Methodology**: Neural scalar potential with state-dependent convex weights; Student-t residual for uncertainty; annealed IDM prior during training.
    - **Outcome**: Manuscript in preparation (co-author).

- **Cooperative Driving Automation System for Urban Areas (MMITSS)** *Sept.2021 - Jun.2026*
  Multiple Projects, Funded by USDOT FHWA, Michigan DOT (via WSP), and Leidos Inc.
  - **Overview**: A vehicle-infrastructure cooperative (V2X) system that connects traffic signals, roadside units, and vehicles, so that intersections can grant signal priority to transit vehicles and guide connected and automated vehicles; built from software through hardware-in-the-loop testing to field deployment, co-simulation, and a test vehicle.
  - **Part 1: Smart Intersections: Paving the Way for a National CAV Deployment** – USDOT FHWA; *Lead Developer and Researcher* *Oct.2023 - Jun.2026*
    - Redeveloped the Multi-Modal Intelligent Traffic Signal System (MMITSS), an open-source connected-vehicle signal priority system (roadside and vehicle-side processors, SAE J2735 messages). [GitHub]
    - Built deployment configurations for Mcity and the Ellsworth Road corridor, set up hardware-in-the-loop testing, and led field testing and data processing; transit signal priority has served Route 6 transit buses on the corridor since August 2025. [GitHub]
    - Maintain road-side and vehicle-side applications as public Docker images (over 700 pulls on Docker Hub as of October 2026); MMITSS is the class demo of Purdue CCE 565 Traffic Operations and Control.
  - **Part 2: Development of Transit Signal Priority and EV Preemption for the Intelligent Woodward Corridor Project** – WSP, for Michigan DOT's ATCMTD project on M-1 (Detroit QLine); *Mentor and Developer* *Jan.2024 - Jun.2026*
    - Implemented MMITSS for the QLine streetcar and extended it to the NTCIP 1211 operation protocol; mentored Ph.D. student Yumeng Bai on the follow-on MMITSS development. [GitHub]

- **Part 3: Integration CDASim with MMITSS and Performance Analysis** – Leidos, for USDOT FHWA's CARMA cooperative driving automation program; *Developer* *Sept.2024 - Apr.2025*
  - Integrated MMITSS into CARMA-CARLA-SUMO co-simulation, with new features on time synchronization, message brokers, and parallel processing; 12 pull requests merged into three FHWA repositories.
- **Part 4: Signal V2X-Based Versatile Connected Autonomous Vehicle** – *Core Team Member*
  - Full-stack, closed-loop infrastructure-to-AV system: connected the signal controller to the roadside V2X unit (SAE J2735), and developed and deployed the vehicle-side **signal-aware speed advisory** algorithm on a **real vehicle**; 2024 Mcity 2.0 Challenge.

- **Safety Instinct for Physical AI: A CBF-Based Safety Cerebellum for Quadruped Robots** *Feb.2026 - Apr.2026*
  Independent Research | Internship | SaferDrive AI(Inchor)
  - **Overview**: An internal exploration of a "safety instinct" for Physical AI – a safety cerebellum for a Boston Dynamics Spot robot, built in **NVIDIA Isaac Sim (Omniverse)**.
  - **Methodology**: Trained a **reinforcement learning (RL)** waypoint-following controller for Spot, then layered a **Control Barrier Function (CBF)** safety filter beneath it – the learned policy handles navigation while the CBF layer enforces safety.
  - **Contribution**: Enabled Spot to maintain a **dynamic safety boundary around moving obstacles** while traveling between waypoints; project conceived and completed independently.
  - **Tech Stack**: NVIDIA Isaac Sim (Omniverse), Reinforcement Learning, Control Barrier Functions

Technical Demonstrations

- **IEEE Intelligent Vehicles Symposium (IV 2026)**, Mcity, Ann Arbor, MI: MMITSS transit signal priority demonstration for symposium attendees and USDOT officials. Jun.2026
- **ITS America Conference & Expo 2026**, Detroit, MI: MMITSS phase-2 demonstration for Michigan DOT and other state DOT officials. Jun.2026
- **UMTRI 60th Anniversary Symposium**, Mcity, Ann Arbor, MI: live demonstration of V2X-enabled transit signal priority with a transit bus. Oct.2025

Skills

- **Programming Languages:** Python, C++, MATLAB, JavaScript
- **Data Science & Machine Learning:** PyTorch, CUDA, Hugging Face
- **Cloud Technologies:** AWS, SSH
- **DevOps & Version Control:** Linux, Docker, Conda
- **Cooperation Platform & Project Management:** Jira, Slack, GitHub
- **Other Tools & Technologies:** Microsoft Office, Auto-CAD, ArcGIS
- **Research Skills:** Critical Thinking, Research Talk, Comprehensive Writing, Literature Review

Honors and Awards

- **Honorable Mention** *2018*
  Mathematical Contest in Modeling of USA [🌐]
- **Silver Award** *2018*
  China College Students' "Internet+" Innovation and Entrepreneurship Competition (team leader)

Additional Information

Languages: English (Proficiency), Chinese Mandarin (Native)
Interests: Soccer, Skateboarding, Range and Gaming

References

1. **Yiheng Feng**
   Associate Professor, Civil and Construction Engineering
   Lyles School of Civil and Construction Engineering, Purdue University
   Relationship: Ph.D. Advisor

2. **Henry Liu**
   Bruce D. Greenshields Collegiate Professor of Engineering
   Director, University of Michigan Transportation Research Institute (UMTRI)
   Department of Civil and Environmental Engineering, University of Michigan
   Relationship: Internship Supervisor, Inchor (SaferDrive AI); Ph.D. Committee Member

3. **Ziran Wang**
   Assistant Professor, Civil and Construction Engineering
   Lyles School of Civil and Construction Engineering, Purdue University
   Relationship: Ph.D. Committee Member

Contact details for references available upon request.